

Indian Institute of Technology Jodhpur

Subject: Time Series Analysis Problem Sheet-IV

Instructions: All text files(.txt) provided in each question are on following link:

<https://faculty.chicagobooth.edu/ruey-s-tsay/research/analysis-of-financial-time-series-3rd-edition>

Question: Answer the following questions.

1. Consider the dataset **m-msft8608.txt** which contains the monthly simple returns of **Microsoft stock**. Plot the log return series and check whether there is an ARCH effect or not.
2. Consider the dataset **m-c8608.txt** which contains the monthly simple returns of **Citi-group stock**. Based on AIC, which is the best fitted model among ARCH(1), ARCH(2) and ARCH(3) with Gaussian innovation? Write down the model equation for best fitted model.
3. Consider the dataset **m-gm3dx7508.txt** which contains the monthly simple returns of **IBM stock, VW, EW, and S&P**. Out of these simple returns consider only monthly simple returns of **IBM stock** given in column **gm**. Based on BIC, which is the best fitted model among ARCH(2), ARCH(3) and ARCH(4) with student-t innovation? Write down the model equation for best fitted model.
4. Consider the dataset **m-gmsp5008.txt** which contains the monthly simple returns of **GM stock & SP500**. Out of these simple returns consider only monthly simple returns of **SP500** given in column **sp**. Fit ARCH(1) model with skew student-t innovation and check whether square of standardized residuals are iid noise or not.
5. Consider the dataset **m-3m4608.txt** which contains the monthly simple returns of **3M stock**. Fit GARCH(1,1) model with student-t innovation and check whether square of standardized residuals are iid noise or not.
6. Consider the dataset **m-intc7308.txt** which contains the monthly simple returns of **Intel stock**. Fit an appropriate GARCH model.
7. Consider the dataset **m-ge2608.txt** which contains the monthly returns of **GE stock**. Fit GARCH(2,1) model with Gaussian, student-t and skew student-t innovations. Plot the volatility of all three models and find correlation between them. Also, plot ACF and PACF of square of standardized residuals of each model.